Task 1:

```
bodike@bodike-chaithali:~/Home$ ls
bodike@bodike-chaithali:~/Home$ mkdir students_marks
bodike@bodike-chaithali:~/Home$ ls
students_marks
bodike@bodike-chaithali:~/Home$ cd students_marks/
bodike@bodike-chaithali:~/Home/students_marks$ ls
bodike@bodike-chaithali:~/Home/students_marks$ touch marksheet.txt
bodike@bodike-chaithali:~/Home/students_marks$ ls
marksheet.txt
bodike@bodike-chaithali:~/Home/students_marks$ cat marksheet.txt
bodike@bodike-chaithali:~/Home/students_marks$ cat > marksheet.txt << EOF
> 101, Alice, Math, 85
> 102, Bob, Math, 72
> 103, Charlie, Math, 88
> 104, David, Math, 66
> 105, Eve, Math, 93
> 106, Frank, Math, 58
> 107, Grace, Math, 81
> 108,Hannah, Math, 75
> 109, Ivan, Math, 69
> 110, Jack, Math, 90
> 101, Alice, Physics, 78
> 102, Bob, Physics, 65
> 103, Charlie, Physics, 82
> 104, David, Physics, 74
> 105, Eve, Physics, 89
> 106, Frank, Physics, 60
> 107, Grace, Physics, 84
> 108, Hannah, Physics, 68
> 109, Ivan, Physics, 71
> 110, Jack, Physics, 86
> 101, Jack, Chemistry, 91
> 102, Bob, Chemistry, 70
> 103, Charlie, Chemistry, 90
> 104, David, Chemistry, 69
> 105, Eve, Chemistry, 95
> 106, Frank, Chemistry, 62
> 107, Grace, Chemistry, 79
```

```
> 107, Grace, Chemistry, 79
> 108, Hannah, Chemistry, 73
> 109, Ivan, Chemistry, 74
> 110, Jack, Chemistry, 88
> EOF
bodike@bodike-chaithali:~/Home/students_marks$ cat marksheet.txt
101, Alice, Math, 85
102, Bob, Math, 72
103, Charlie, Math, 88
104, David, Math, 66
105, Eve, Math, 93
106, Frank, Math, 58
107, Grace, Math, 81
108,Hannah, Math, 75
109, Ivan, Math, 69
110, Jack, Math, 90
101, Alice, Physics, 78
102, Bob, Physics, 65
103, Charlie, Physics, 82
104, David, Physics, 74
105, Eve, Physics, 89
106, Frank, Physics, 60
107, Grace, Physics, 84
108, Hannah, Physics, 68
109, Ivan, Physics, 71
110, Jack, Physics, 86
101, Jack, Chemistry, 91
102, Bob, Chemistry, 70
103, Charlie, Chemistry, 90
104, David, Chemistry, 69
105, Eve, Chemistry, 95
106, Frank, Chemistry, 62
107, Grace, Chemistry, 79
108, Hannah, Chemistry, 73
109, Ivan, Chemistry, 74
110, Jack, Chemistry, 88
bodike@bodike-chaithali:~/Home/students_marks$
```

Q1)

```
105, Eve, Math, 93
106, Frank, Math, 58
107, Grace, Math, 81
108,Hannah, Math, 75
109, Ivan, Math, 69
110, Jack, Math, 90
101, Alice, Physics, 78
102, Bob, Physics, 65
103, Charlie, Physics, 82
104, David, Physics, 74
105, Eve, Physics, 89
106, Frank, Physics, 60
107, Grace, Physics, 84
108, Hannah, Physics, 68
109, Ivan, Physics, 71
110, Jack, Physics, 86
101, Jack, Chemistry, 91
102, Bob, Chemistry, 70
103, Charlie, Chemistry, 90
104, David, Chemistry, 69
105, Eve, Chemistry, 95
106, Frank, Chemistry, 62
107, Grace, Chemistry, 79
108, Hannah, Chemistry, 73
109, Ivan, Chemistry, 74
110, Jack, Chemistry, 88
bodike@bodike-chaithali:~/Home/students_marks$ sed -n '10,20p' marksheet.txt
110, Jack, Math, 90
101, Alice, Physics, 78
102, Bob, Physics, 65
103, Charlie, Physics, 82
104, David, Physics, 74
105, Eve, Physics, 89
106, Frank, Physics, 60
107, Grace, Physics, 84
108, Hannah, Physics, 68
109, Ivan, Physics, 71
110, Jack, Physics, 86
```

Q2)

```
bodike@bodike-chaithali:~/Home/students_marks$ sed -i 's/,/::/g' marksheet.txt
bodike@bodike-chaithali:~/Home/students_marks$ cat marksheet.txt
101:: Alice:: Math:: 85
102:: Bob:: Math:: 72
103:: Charlie:: Math:: 88
104:: David:: Math:: 66
105:: Eve:: Math:: 93
106:: Frank:: Math:: 58
107:: Grace:: Math:: 81
108::Hannah:: Math:: 75
109:: Ivan:: Math:: 69
110:: Jack:: Math:: 90
101:: Alice:: Physics:: 78
102:: Bob:: Physics:: 65
103:: Charlie:: Physics:: 82
104:: David:: Physics:: 74
105:: Eve:: Physics:: 89
106:: Frank:: Physics:: 60
107:: Grace:: Physics:: 84
108:: Hannah:: Physics:: 68
109:: Ivan:: Physics:: 71
110:: Jack:: Physics:: 86
101:: Alice:: Chemistry:: 91
102:: Bob:: Chemistry:: 70
103:: Charlie:: Chemistry:: 90
104:: David:: Chemistry:: 69
105:: Eve:: Chemistry:: 95
106:: Frank:: Chemistry:: 62
107:: Grace:: Chemistry:: 79
108:: Hannah:: Chemistry:: 73
109:: Ivan:: Chemistry:: 74
110:: Jack:: Chemistry:: 88
bodike@bodike-chaithali:~/Home/students_marks$
```

Q3)

```
107:: Grace:: Math:: 81
108::Hannah:: Math:: 75
109:: Ivan:: Math:: 69
110:: Jack:: Math:: 90
101:: Alice:: Physics:: 78
102:: Bob:: Physics:: 65
103:: Charlie:: Physics:: 82
104:: David:: Physics:: 74
105:: Eve:: Physics:: 89
106:: Frank:: Physics:: 60
107:: Grace:: Physics:: 84
108:: Hannah:: Physics:: 68
109:: Ivan:: Physics:: 71
110:: Jack:: Physics:: 86
101:: Jack:: Chemistry:: 91
102:: Bob:: Chemistry:: 70
103:: Charlie:: Chemistry:: 90
104:: David:: Chemistry:: 69
105:: Eve:: Chemistry:: 95
106:: Frank:: Chemistry:: 62
107:: Grace:: Chemistry:: 79
108:: Hannah:: Chemistry:: 73
109:: Ivan:: Chemistry:: 74
110:: Jack:: Chemistry:: 88
bodike@bodike-chaithali:~/Home/students_marks$ awk -F "::" '{marks[$1]+=$4;name[$1]=$2;} END { for(i in marks) print i "," name[i] "," marks[i
] ",300" }' marksheet.txt > total_marks.txt
bodike@bodike-chaithali:~/Home/students_marks$ cat total_marks.txt
102, Bob,207,300
109, Ivan,214,300
107, Grace,244,300
104, David,209,300
103, Charlie,260,300
105, Eve,277,300
110, Jack,264,300
101, Jack,254,300
108, Hannah,216,300
106, Frank,180,300
bodike@bodike-chaithali:~/Home/students_marks$
```

Q4)

```
bodike@bodike-chaithali:~/Home/students_marks$ awk -F "," ' \
{ \
  percent = ($3/300)*100; \
  grade = (percent >= 90) ? "A" : \
          (percent >= 80) ? "B" : \
          (percent >= 70) ? "C" : \
          (percent >= 60) ? "D" : "E"; \
  print $1 "::" $2 "::" grade; \
}
' total_marks.txt
102:: Bob::D
109:: Ivan::C
107:: Grace::B
104:: David::D
103:: Charlie::B
105:: Eve::A
110:: Jack::B
101:: Jack::B
108:: Hannah::C
106:: Frank::D
bodike@bodike-chaithali:~/Home/students_marks$
```

Q5)

```
bodike@bodike-chaithali:~/Home/students_marks$ sed -i '/:: Frank::/d' marksheet.txt
bodike@bodike-chaithali:~/Home/students_marks$ cat marksheet.txt
101:: Alice:: Math:: 85
102:: Bob:: Math:: 72
103:: Charlie:: Math:: 88
104:: David:: Math:: 66
105:: Eve:: Math:: 93
107:: Grace:: Math:: 81
108::Hannah:: Math:: 75
109:: Ivan:: Math:: 69
110:: Jack:: Math:: 90
101:: Alice:: Physics:: 78
102:: Bob:: Physics:: 65
103:: Charlie:: Physics:: 82
104:: David:: Physics:: 74
105:: Eve:: Physics:: 89
107:: Grace:: Physics:: 84
108:: Hannah:: Physics:: 68
109:: Ivan:: Physics:: 71
110:: Jack:: Physics:: 86
101:: Jack:: Chemistry:: 91
102:: Bob:: Chemistry:: 70
103:: Charlie:: Chemistry:: 90
104:: David:: Chemistry:: 69
105:: Eve:: Chemistry:: 95
107:: Grace:: Chemistry:: 79
108:: Hannah:: Chemistry:: 73
109:: Ivan:: Chemistry:: 74
110:: Jack:: Chemistry:: 88
bodike@bodike-chaithali:~/Home/students_marks$
```

```
bodike@bodike-chaithali:~/Home/students_marks$ sed -i '/Frank/d' total_marks.txt
bodike@bodike-chaithali:~/Home/students_marks$ cat total_marks.txt
102, Bob,207,300
109, Ivan,214,300
107, Grace,244,300
104, David,209,300
103, Charlie,260,300
105, Eve,277,300
110, Jack,264,300
101, Jack,254,300
108, Hannah,216,300
bodike@bodike-chaithali:~/Home/students_marks$
```

Q6)

```
bodike@bodike-chaithali:~/Home/students_marks$ sort -t',' -k3,3nr total_marks.txt -o total_marks.txt
bodike@bodike-chaithali:~/Home/students_marks$ cat total_marks.txt
105, Eve,277,300
110, Jack,264,300
103, Charlie,260,300
101, Jack,254,300
107, Grace,244,300
108, Hannah,216,300
109, Ivan,214,300
104, David,209,300
102, Bob,207,300
bodike@bodike-chaithali:~/Home/students_marks$
```

Q7)

```
bodike@bodike-chaithali:~/Home/students_marks$ sed -i 's/Eve/Eva/g' total_marks.txt
bodike@bodike-chaithali:~/Home/students_marks$ cat total_marks.txt
105, Eva,277,300
110, Jack,264,300
103, Charlie,260,300
101, Jack,254,300
107, Grace,244,300
108, Hannah,216,300
109, Ivan,214,300
104, David,209,300
102, Bob,207,300
bodike@bodike-chaithali:~/Home/students_marks$
```

```
bodike@bodike-chaithali:~/Home/students_marks$ sed -i 's/Eve/Eva/g' marksheet.txt
bodike@bodike-chaithali:~/Home/students_marks$ cat marksheet.txt
101:: Alice:: Math:: 85
102:: Bob:: Math:: 72
103:: Charlie:: Math:: 88
104:: David:: Math:: 66
105:: Eva:: Math:: 93
107:: Grace:: Math:: 81
108::Hannah:: Math:: 75
109:: Ivan:: Math:: 69
110:: Jack:: Math:: 90
101:: Alice:: Physics:: 78
102:: Bob:: Physics:: 65
103:: Charlie:: Physics:: 82
104:: David:: Physics:: 74
105:: Eva:: Physics:: 89
107:: Grace:: Physics:: 84
108:: Hannah:: Physics:: 68
109:: Ivan:: Physics:: 71
110:: Jack:: Physics:: 86
101:: Jack:: Chemistry:: 91
102:: Bob:: Chemistry:: 70
103:: Charlie:: Chemistry:: 90
104:: David:: Chemistry:: 69
105:: Eva:: Chemistry:: 95
107:: Grace:: Chemistry:: 79
108:: Hannah:: Chemistry:: 73
109:: Ivan:: Chemistry:: 74
110:: Jack:: Chemistry:: 88
bodike@bodike-chaithali:~/Home/students_marks$
```

Task 2:

```
bodike@bodike-chaithali:~/Home$ ls
students_marks
bodike@bodike-chaithali:~/Home$ mkdir dir2
bodike@bodike-chaithali:~/Home$ ls
dir2  students_marks
bodike@bodike-chaithali:~/Home$ cd dir2/
bodike@bodike-chaithali:~/Home/dir2$ ls
bodike@bodike-chaithali:~/Home/dir2$ touch student_id.txt
bodike@bodike-chaithali:~/Home/dir2$ ls
student_id.txt
bodike@bodike-chaithali:~/Home/dir2$ cat student_id.txt
bodike@bodike-chaithali:~/Home/dir2$ cat > student_id.txt << EOF
> 1002::Sukla
> 1003::Devi
> 1005::Nath
> EOF
bodike@bodike-chaithali:~/Home/dir2$ cat student_id.txt
1002::Sukla
1003::Devi
1005::Nath
bodike@bodike-chaithali:~/Home/dir2$ touch student_age.txt
bodike@bodike-chaithali:~/Home/dir2$ ls
student_age.txt  student_id.txt
bodike@bodike-chaithali:~/Home/dir2$ cat student_age.txt
bodike@bodike-chaithali:~/Home/dir2$ cat > student_age.txt << EOF
> 1005::23
> 1002::21
> 1003::18
> EOF
bodike@bodike-chaithali:~/Home/dir2$ cat student_age.txt
1005::23
1002::21
1003::18
bodike@bodike-chaithali:~/Home/dir2$
```

Q1)

```
bodike@bodike-chaithali:~/Home/dir2$ awk -F'::' 'NR==FNR {a[$1]=$2; next} {print $1 "::" a[$1] "::" $2}' student_id.txt student_age.txt > id_n
ame_age.txt
bodike@bodike-chaithali:~/Home/dir2$ cat id_name_age.txt
1005::Nath::23
1002::Sukla::21
1003::Devi::18
bodike@bodike-chaithali:~/Home/dir2$
```

Q2)

```
bodike@bodike-chaithali:~/Home/dir2$ awk -F'::' '{ print $3 "::" $0 }' id_name_age.txt | sort -n | cut -d':' -f3- > temp && mv temp id_name_ag
e.txt
bodike@bodike-chaithali:~/Home/dir2$ cat id_name_age.txt
1003::Devi::18
1002::Sukla::21
1005::Nath::23
bodike@bodike-chaithali:~/Home/dir2$
```

Q3)

```
bodike@bodike-chaithali:~/Home/dir2$ awk -F'::' '{ print $3 "::" $0 }' id_name_age.txt | sort -n | cut -d':' -f3- > temp && mv temp id_name_ag
e.txt
bodike@bodike-chaithali:~/Home/dir2$ cat id_name_age.txt
1003::Devi::18
1002::Sukla::21
1005::Nath::23
bodike@bodike-chaithali:~/Home/dir2$ awk -F'::' '{ print $1 "::" $3 }' id_name_age.txt
1003::18
1002::21
1005::23
bodike@bodike-chaithali:~/Home/dir2$
```

Q4)

```
bodike@bodike-chaithali:~/Home/dir2$ ls
id_name_age.txt  student_age.txt  student_id.txt
bodike@bodike-chaithali:~/Home/dir2$ cp student_id.txt student_id_new.txt
bodike@bodike-chaithali:~/Home/dir2$ ls
id_name_age.txt  student_age.txt  student_id_new.txt  student_id.txt
bodike@bodike-chaithali:~/Home/dir2$ cat student_id_new.txt
1002::Sukla
1003::Devi
1005::Nath
bodike@bodike-chaithali:~/Home/dir2$ sed -i 's/1002::Sukla/1023::Sukla/' student_id_new.txt
bodike@bodike-chaithali:~/Home/dir2$ cat student_id_new.txt
1023::Sukla
1003::Devi
1005::Nath
bodike@bodike-chaithali:~/Home/dir2$ diff -y student_id.txt student_id_new.txt
1002::Sukla                                                     | 1023::Sukla
1003::Devi                                                        1003::Devi
1005::Nath                                                        1005::Nath
bodike@bodike-chaithali:~/Home/dir2$
```

Q5)

```
bodike@bodike-chaithali:~/Home/dir2$ cd ..
bodike@bodike-chaithali:~/Home$ ls
dir2  students_marks
bodike@bodike-chaithali:~/Home$ grep -w "23" dir2/*
dir2/id_name_age.txt:1005::Nath::23
dir2/student_age.txt:1005::23
bodike@bodike-chaithali:~/Home$
```

Q6)

```
bodike@bodike-chaithali:~/Home$ cd dir2
bodike@bodike-chaithali:~/Home/dir2$ ls
id_name_age.txt  student_age.txt  student_id_new.txt  student_id.txt
bodike@bodike-chaithali:~/Home/dir2$ awk -F'::' '{ print $2 "::" $0 }' student_id_new.txt | sort | cut -d':' -f3- > temp && mv temp student_id
_new.txt
bodike@bodike-chaithali:~/Home/dir2$ cat student_id_new.txt
1003::Devi
1005::Nath
1023::Sukla
bodike@bodike-chaithali:~/Home/dir2$
```

Q7)

```
bodike@bodike-chaithali:~/Home/dir2$ cmp -s student_id.txt student_id_new.txt && echo "Same" || echo "Different"
Different
bodike@bodike-chaithali:~/Home/dir2$ cat student_id.txt
1002::Sukla
1003::Devi
1005::Nath
bodike@bodike-chaithali:~/Home/dir2$ cat student_id_new.txt
1003::Devi
1005::Nath
1023::Sukla
bodike@bodike-chaithali:~/Home/dir2$
```